

DEPLOYING MACHINE LEARNING MODELS IN PRODUCTION

05 optional build train and deploy an xgboost model on cloud ai platform C4 W1 Optional Lab 1

Source: 05_optional-build-train-and-deploy-an-xgboost-model-on-cloud-ai-platform_C4_W1_Optional_Lab_1.md

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Specialization - machine-learning-engineering-for-production-public/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md at mai..." /><meta
property="og:image:width" content="1200" /><meta property="og:image:height" content="600"
/><meta property="og:site_name" content="GitHub" /><meta property="og:type"
content="object" /><meta property="og:title" content="machine-learning-engineering-for-
production-public/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md at main · https-
deeplearning-ai/machine-learning-engineering-for-production-public" /><meta
property="og:url" content="https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public" /><meta property="og:description" content="Public repo
for DeepLearning.AI MLEP Specialization - machine-learning-engineering-for-production-
public/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md at mai..." />

  <meta name="hostname" content="github.com">

  <meta name="expected-hostname" content="github.com">

```

```
<meta name="enabled-features"
content="TURBO_EXPERIMENT_RISKY, IMAGE_METRIC_TRACKING, GEOJSON_AZURE_MAPS, SUDO_MODAL_FOCUS">
```

```
<meta data-hydrostats="publish">
```

```
<link rel="canonical" href="https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md" data-turbo-transient>
```

```
<div class="position-relative js-header-wrapper ">
  <a href="#start-of-content" class="px-2 py-4 color-bg-accent-emphasis color-fg-on-
emphasis show-on-focus js-skip-to-content">Skip to content</a>
  <span data-view-component="true" class="progress-pjax-loader Progress position-fixed
width-full">
<span style="width: 0%;" data-view-component="true" class="Progress-item progress-pjax-
loader-bar left-0 top-0 color-bg-accent-emphasis"></span>
```

```
<script crossorigin="anonymous" defer="defer" type="application/javascript"
src="https://github.githubassets.com/assets/vendors-node_modules_github_remote-
form_dist_index_js-node_modules_github_memoize_dist_esm_in-687f35-d131f0b6de8e.js">
</script>
```

Toggle navigation



```

<div class="flex-1">
  <a href="/signup?
ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F%3Cuser-name%3E%2F%3Crepo-
name%3E%2Fblob%2Fshow&source=header-repo"
    class="d-inline-block d-lg-none flex-order-1 f5 no-underline border color-border-
default rounded-2 px-2 py-1 color-fg-inherit"
    data-hydro-click="{
"event_type":"authentication.click","payload":"
"location_in_page":"site
header","repository_id":"null,"auth_type":"SIGN_UP","
"originating_url":"https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md","user_id":nu
ll}" data-hydro-click-
hmac="54741315ea6afc6c42a65c564e4adf174b717ea1b7eb805cd428b6f526a77739"
  >
    Sign up
  </a>
</div>

<div class="flex-1 flex-order-2 text-right">
  <button aria-label="Toggle navigation" aria-expanded="false" type="button" data-view-
component="true" class="js-details-target Button--link Button--medium Button d-lg-none
color-fg-inherit p-1">    <span class="Button-content">
    <span class="Button-label"><div class="HeaderMenu-toggle-bar rounded my-1"></div>
      <div class="HeaderMenu-toggle-bar rounded my-1"></div>
      <div class="HeaderMenu-toggle-bar rounded my-1"></div></span>
    </span>
</div>

```

```

<div class="HeaderMenu--logged-out p-responsive height-fit position-lg-relative d-lg-flex
flex-column flex-auto pt-7 pb-4 top-0">
  <div class="header-menu-wrapper d-flex flex-column flex-self-end flex-lg-row flex-justify-
between flex-auto p-3 p-lg-0 rounded rounded-lg-0 mt-3 mt-lg-0">
    <nav class="mt-0 px-3 px-lg-0 mb-3 mb-lg-0" aria-label="Global">
      <ul class="d-lg-flex list-style-none">
        <li class="HeaderMenu-item position-relative flex-wrap flex-justify-between flex-
items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-container js-
header-menu-item">
          <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0 px-
lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-target"
aria-expanded="false">
            Product
            <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-chevron-down HeaderMenu-icon
ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8 8.93l3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>

```



Actions

Automate any workflow



Packages

Host and manage packages



Security

Find and fix vulnerabilities



Codespaces

Instant dev environments



Copilot

Write better code with AI



Code review

Manage code changes



Issues

Plan and track work



Discussions

Collaborate outside of code

```

    </ul>
  </div>
  <div class="px-lg-4">
    <span class="d-block h4 color-fg-default my-1" id="product-explore-
heading">Explore</span>
    <ul class="list-style-none f5" aria-labelledby="product-explore-heading">
      <li>

```

All features

```
<li>
```

Documentation

```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-
view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon
color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>

```

```
<li>
```

GitHub Skills

```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-
view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon
color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>

```

```
<li>
```

[Blog](#)

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>
```

```
</ul>
</div>
</div>
```

```
<li class="HeaderMenu-item position-relative flex-wrap flex-justify-between
flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-container
js-header-menu-item">
  <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0
px-lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-
target" aria-expanded="false">
    Solutions
    <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-chevron-down HeaderMenu-
icon ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8 8.939l3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>
```

For

- [Enterprise](#)

```
<li>
```

[Teams](#)

```
<li>
```

[Startups](#)

```
<li>
```

Education

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
<div class="border-bottom pb-3 mb-3">
  <span class="d-block h4 color-fg-default my-1" id="solutions-by-solution-
heading">By Solution</span>
  <ul class="list-style-none f5" aria-labelledby="solutions-by-solution-
heading">
    <li>
```

CI/CD & Automation

```
<li>
```

DevOps

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
<li>
```

DevSecOps

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
<div class="">
  <span class="d-block h4 color-fg-default my-1" id="solutions-case-studies-
heading">Case Studies</span>
  <ul class="list-style-none f5" aria-labelledby="solutions-case-studies-
heading">
    <li>
```

Customer Stories

```
<li>
```

Resources

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
</div>
```

```

<li class="HeaderMenu-item position-relative flex-wrap flex-justify-between
flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-
container js-header-menu-item">
  <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0
px-lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-
target" aria-expanded="false">
    Open Source
    <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16"
version="1.1" width="16" data-view-component="true" class="octicon octicon-chevron-
down HeaderMenu-icon ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8 8.939l3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>

```

```

■ <div>
  <div class="color-fg-default h4">GitHub Sponsors</div>
  Fund open source developers
</div>

```

```

  </ul>
</div>
<div class="border-bottom pb-3 mb-3">
  <ul class="list-style-none f5" >
    <li>

```

```

<div>
  <div class="color-fg-default h4">The ReadME Project</div>
  GitHub community articles
</div>

```

```

  </ul>
</div>
<div class="">
  <span class="d-block h4 color-fg-default my-1" id="open-source-
repositories-heading">Repositories</span>
  <ul class="list-style-none f5" aria-labelledby="open-source-repositories-
heading">
    <li>

```

Topics

```
<li>
```

Trending

```
<li>
```

Collections

```

    </ul>
  </div>
</div>

```

```

    <li class="HeaderMenu-item position-relative flex-wrap flex-justify-
between flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-
details-container js-header-menu-item">
<a class="HeaderMenu-link no-underline px-0 px-lg-2 py-3 py-lg-2 d-block d-lg-
inline-block" data-analytics-event="{&quot;category&quot;:&quot;Header menu top
item (logged out)&quot;;&quot;action&quot;:&quot;click to go to
Pricing&quot;;&quot;label&quot;:&quot;ref_cta:Pricing&quot;}"
href="/pricing">Pricing</a>

```

```

    </ul>
  </nav>

  <div class="d-lg-flex flex-items-center px-3 px-lg-0 mb-3 mb-lg-0 text-center
text-lg-left">
    <div class="d-lg-flex min-width-0 mb-2 mb-lg-0">

```

```

placeholder="Search" data-unscoped-placeholder="Search
GitHub" data-scoped-placeholder="Search" autocapitalize="off" role="combobox" aria-
haspopup="listbox" aria-expanded="false" aria-autocomplete="list" aria-
controls="jump-to-results" aria-label="Search" data-jump-to-suggestions-
path="/_graphql/GetSuggestedNavigationDestinations" spellcheck="false"
autocomplete="off" > <input type="hidden" data-csrf="true" class="js-data-jump-to-
suggestions-path-csrf"
value="VUIz3+t1V4vxrbGWggpJPJxD5T6RWpPXDkn0W5Nbs1NcwE0RPxiYpSeRYQ93pRZKQH3/F0UNuXSe
wUYCKjfQbg==" /> <input type="hidden" class="js-site-search-type-field" name="type"
> <svg xmlns="http://www.w3.org/2000/svg" width="22" height="20" aria-hidden="true"
class="mr-1 header-search-key-slash"><path fill="none" stroke="#979A9C" opacity=".4"
d="M3.5.5h12c1.7 0 3 1.3 3 3v13c0 1.7-1.3 3-3 3h-12c-1.7 0-3-1.3-3-3v-13c0-1.7 1.3-
3 3-3z"></path><path fill="#979A9C" d="M11.8 6L8 15.1h-.9L10.8 6h1z"></path></svg>
<div class="Box position-absolute overflow-hidden d-none jump-to-suggestions js-
jump-to-suggestions-container">

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```

- No suggested jump to results
- 📄 🗃️ 🔍

```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this organization">
    In this organization
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

```

```

  </div>
</label>

```

```

</div>

<div class="position-relative mr-lg-3 d-lg-inline-block">
  <a href="/login?return_to=https%3A%2F%2Fgithub.com%2Fhttps-deeplearning-
ai%2Fmachine-learning-engineering-for-production-
public%2Fblob%2Fmain%2Fcourse4%2Fweek1-ungraded-
labs%2FC4_W1_Optional_Lab_1_XGBoost_CAIP%2FC4_W1_Optional_Lab_1.md"
    class="HeaderMenu-link HeaderMenu-link--sign-in flex-shrink-0 no-
underline d-block d-lg-inline-block border border-lg-0 rounded rounded-lg-0 p-2 p-
lg-0"
    data-hydro-click="{
  "event_type":"authentication.click";"payload":
  "location_in_page":"site header
menu";"repository_id":null,"auth_type":"SIGN_UP"
  "originating_url":"https://github.com/https-deeplearning-
ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-
ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id
  ":null}"" data-hydro-click-
  hmac="64b4081eadf9f5bce2b2243bf763a9331e35748ce2261b74b90230c06babee37"
  data-ga-click="(Logged out) Header, clicked Sign in, text:sign-in">
    Sign in
  </a>
</div>

  <a href="/signup?
ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F%3Cuser-
name%3E%2F%3Crepo-name%3E%2Fblob%2Fshow&source=header-
repo&source_repo=https-deeplearning-ai%2Fmachine-learning-engineering-for-
production-public"
    class="HeaderMenu-link HeaderMenu-link--sign-up flex-shrink-0 d-none d-
lg-inline-block no-underline border color-border-default rounded px-2 py-1"
    data-hydro-click="{
  "event_type":"authentication.click";"payload":
  "location_in_page":"site header
menu";"repository_id":null,"auth_type":"SIGN_UP"
  "originating_url":"https://github.com/https-deeplearning-
ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-
ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id
  ":null}"" data-hydro-click-
  hmac="64b4081eadf9f5bce2b2243bf763a9331e35748ce2261b74b90230c06babee37"
  data-analytics-event="{"category":"Sign
up";"action":"click to sign up for
account";"label":"ref_page:/&lt;user-name&gt;/&lt;repo-
name&gt;/blob/show;ref_cta:Sign up;ref_loc:header logged out"}"
  >
    Sign up
  </a>
</div>
</div>
</div>

```

```
<div hidden="hidden" data-view-component="true" class="js-stale-session-flash flash flash-warn mb-3">
```

```
  <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-alert">
    <path d="M6.457 1.047c.659-1.234 2.427-1.234 3.086 0l6.082 11.378A1.75 1.75 0 0 1 14.082 15H1.918a1.75 1.75 0 0 1-1.543-2.575Zm1.763.707a.25.25 0 0 0-.44 0L1.698 13.132a.25.25 0 0 0 .22.368h12.164a.25.25 0 0 0 .22-.368Zm.53 3.996v2.5a.75.75 0 0 1-1.5 0v-2.5a.75.75 0 0 1 1.5 0ZM9 11a1 1 0 1 1-2 0 1 1 0 0 1 2 0Z"></path>
```

```
<button class="flash-close js-flash-close" type="button" aria-label="Close">
  <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-x">
    <path d="M3.72 3.72a.75.75 0 0 1 1.06 0L8 6.94l3.22-3.22a.749.749 0 0 1 1.275.326.749.749 0 0 1-.215.734L9.06 8l3.22 3.22a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215L8 9.06l-3.22 3.22a.751.751 0 0 1-1.042-.018.751.751 0 0 1-.018-1.042L6.94 8 3.72 4.78a.75.75 0 0 1 0-1.06Z"></path>
```

```
<div id="js-flash-container" data-turbo-replace>
```

```
<include-fragment class="js-notification-shelf-include-fragment" data-base-src="https://github.com/notifications/beta/shelf"></include-fragment>
```

```
<div class="d-flex flex-wrap flex-justify-end mb-3 px-3 px-md-4 px-lg-5" style="gap: 1rem;">
```

```
  <div class="flex-auto min-width-0 width-fit mr-3">
```



```
<span class="author flex-self-stretch" itemprop="author">
  <a class="url fn" rel="author" data-hovercard-type="organization" data-hovercard-url="/orgs/https-deeplearning-ai/hovercard" data-octo-click="hovercard-link-click" data-octo-dimensions="link_type:self" href="/https-deeplearning-ai">
    https-deeplearning-ai
```

/ [machine-learning-engineering-for-production-public](#)

```
<span></span><span class="Label Label--secondary v-align-middle mr-1">Public</span>
```

```
</div>
```

```
<div id="repository-details-container" data-turbo-replace>
  <ul class="pagehead-actions flex-shrink-0 d-none d-md-inline" style="padding: 2px 0;">
```

 [Notifications](#)

 [Fork 1.9k](#)

 [Star 1.5k](#) 

```
</div>
</div>
```

```
<div id="responsive-meta-container" data-turbo-replace>
```

```
<nav data-pjax="#js-repo-pjax-container" aria-label="Repository" data-view-
component="true" class="js-repo-nav js-sidenav-container-pjax js-responsive-underlinenav
overflow-hidden UnderlineNav px-3 px-md-4 px-lg-5">
```

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-code UnderlineNav-octicon
d-none d-sm-inline">
<path d="m11.28 3.22 4.25 4.25a.75.75 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-
1.275-.326.749.749 0 0 1 .215-.734l13.94 8l-3.72-3.72a.749.749 0 0 1 .326-1.275.749.749
0 0 1 .734.215zm-6.56 0a.751.751 0 0 1 1.042.018.751.751 0 0 1 .018 1.042l2.06 8l3.72
3.72a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215l.47 8.53a.75.75 0 0 1 0-1.06z">
</path>
```

## [Code](#)

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-issue-opened UnderlineNav-
octicon d-none d-sm-inline">
<path d="M8 9.5a1.5 1.5 0 1 0 0-3 1.5 1.5 0 0 0 0 3z"></path><path d="M8 0a8 8 0 1 1 0
16A8 8 0 0 1 8 0zM1.5 8a6.5 6.5 0 1 0 13 0 6.5 6.5 0 0 0-13 0z"></path>
```

[Issues 18](#)

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-git-pull-request
UnderlineNav-octicon d-none d-sm-inline">
<path d="M1.5 3.25a2.25 2.25 0 1 1 3 2.122v5.256a2.251 2.251 0 1 1-1.5 0V5.372A2.25 2.25
0 0 1 1.5 3.25zm5.677-.177L9.573.677A.25.25 0 0 1 10 .854V2.5h1A2.5 2.5 0 0 1 13.5
5v5.628a2.251 2.251 0 1 1-1.5 0V5a1 1 0 0 0-1-1h-1v1.646a.25.25 0 0 1-.427.177L7.177
3.427a.25.25 0 0 1 0-.354zM3.75 2.5a.75.75 0 1 0 0 1.5.75.75 0 0 0 0-1.5zm0 9.5a.75.75 0
1 0 0 1.5.75.75 0 0 0 0-1.5zm8.25.75a.75.75 0 1 0 1.5 0 .75.75 0 0 0-1.5 0z"></path>
```

## [Pull requests 2](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-play UnderlineNav-octicon
d-none d-sm-inline">
<path d="M8 0a8 8 0 1 1 0 16A8 8 0 0 1 8 0ZM1.5 8a6.5 6.5 0 1 0 13 0 6.5 6.5 0 0 0-13
0Zm4.879-2.773 4.264 2.559a.25.25 0 0 1 0 .428l-4.264 2.559A.25.25 0 0 1 6
10.559V5.442a.25.25 0 0 1 .379-.215Z"></path>

```

[Actions](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-table UnderlineNav-octicon
d-none d-sm-inline">
<path d="M0 1.75C0 .784.784 0 1.75 0h12.5C15.216 0 16 .784 16 1.75v12.5A1.75 1.75 0 0 1
14.25 16H1.75A1.75 1.75 0 0 1 0 14.25ZM6.5 6.5v8h7.75a.25.25 0 0 0 .25-.25V6.5Zm8-
1.5V1.75a.25.25 0 0 0-.25-.25H6.5V5Zm-13 1.5v7.75c0 .138.112.25.25.25H5v-8ZM5
5V1.5H1.75a.25.25 0 0 0-.25.25V5Z"></path>

```

### [Projects](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-shield UnderlineNav-octicon
d-none d-sm-inline">
<path d="M7.467.133a1.748 1.748 0 0 1 1.066 0l5.25 1.68A1.75 1.75 0 0 1 15 3.48V7c0
1.566-.32 3.182-1.303 4.682-.983 1.498-2.585 2.813-5.032 3.855a1.697 1.697 0 0 1-1.33
0c-2.447-1.042-4.049-2.357-5.032-3.855C1.32 10.182 1 8.566 1 7V3.48a1.75 1.75 0 0 1
1.217-1.667Zm.61 1.429a.25.25 0 0 0-.153 0l-5.25 1.68a.25.25 0 0 0-.174.238V7c0
1.358.275 2.666 1.057 3.86.784 1.194 2.121 2.34 4.366 3.297a.196.196 0 0 0 .154
0c2.245-.956 3.582-2.104 4.366-3.298C13.225 9.666 13.5 8.36 13.5 7V3.48a.251.251 0 0
0-.174-.237l-5.25-1.68ZM8.75 4.75v3a.75.75 0 0 1-1.5 0v-3a.75.75 0 0 1 1.5 0ZM9 10.5a1 1
0 1 1-2 0 1 1 0 0 1 2 0Z"></path>

```

[Security](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-graph UnderlineNav-octicon
d-none d-sm-inline">
<path d="M1.5 1.75V13.5h13.75a.75.75 0 0 1 0 1.5H.75a.75.75 0 0 1-.75-.75V1.75a.75.75 0
0 1 1.5 0Zm14.28 2.53-5.25 5.25a.75.75 0 0 1-1.06 0L7 7.06 4.28 9.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.25-3.25a.75.75 0 0 1 1.06 0L10 7.94l4.72-
4.72a.751.751 0 0 1 1.042.018.751.751 0 0 1 .018 1.042Z"></path>

```

### [Insights](#)

### [Permalink](#)

►  main

[View all tags](#)

## Name already in use

✕

```
<div data-view-component="true"> A tag already exists with the provided branch
name. Many Git commands accept both tag and branch names, so creating this branch may cause
unexpected behavior. Are you sure you want to create this branch?
```

```
</div>
<footer class="Overlay-footer Overlay-footer--alignEnd">
 <button data-close-dialog-id="warn-tag-match-create-branch-dialog" type="button"
data-view-component="true" class="btn"> Cancel
```

Create

### [machine-learning-engineering-for-production-public/course4/week1-ungraded-labs/C4\\_W1\\_Optional\\_Lab\\_1\\_XGBoost\\_CAIP/C4\\_W1\\_Optional\\_Lab\\_1.md](#)

[Go to file](#)

► ...

```
<div id="spoof-warning" class="mt-0 pb-3" hidden aria-hidden>
```

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-
component="true" class="octicon octicon-alert float-left mt-1">
<path d="M6.457 1.047c.659-1.234 2.427-1.234 3.086 0l6.082 11.378A1.75 1.75 0 0 1 14.082
15H1.918a1.75 1.75 0 0 1-1.543-2.575Zm1.763.707a.25.25 0 0 0-.44 0L1.698 13.132a.25.25 0 0
0 .22.368h12.164a.25.25 0 0 0 .22-.368Zm.53 3.996v2.5a.75.75 0 0 1-1.5 0v-2.5a.75.75 0 0 1
1.5 0ZM9 11a1 1 0 1 1-2 0 1 1 0 0 1 2 0Z"></path>
```

```
<div class="overflow-hidden">This commit does not belong to any branch on this repository,
and may belong to a fork outside of the repository.</div>
```

```
<include-fragment src="/https-deeplearning-ai/machine-learning-engineering-for-production-
public/spoofed_commit_check/34af5af3f5b09a52a1537292b0fe14a29ae16b8d" data-test-
selector="spoofed-commit-check"></include-fragment>
```

```
<div class="Box d-flex flex-column flex-shrink-0 mb-3">
```

```
<div class="Box-body d-flex flex-items-center" >
 <div class="Skeleton Skeleton--text col-1"> </div>
 Cannot retrieve contributors at this
time
</div>
```

```
<readme-toc>
```

```
<div data-target="readme-toc.content" class="Box mt-3 position-relative">
```

```
<details
```

```
data-target="readme-toc.trigger" data-menu-hydro-
click="{\"event_type\":\"repository_toc_menu.click\",\"payload\":
{\"target\":\"trigger\",\"repository_id\":313476213,\"originating_url\":\"https://github.com/https-deeplearning-
ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md\";\";\"user_id\":\"null
}}\" data-menu-hydro-click-
hmac=\"ef305c907403ac041ac9d21749f0ba7f383c1b6923c75e8acc0004d49e1c4bc7\" class=\"dropdown
details-reset details-overlay\"
```



```

<div class="SelectMenu-filter">
 <input
 class="SelectMenu-input form-control js-filterable-field"
 id="toc-filter-field"
 type="text"
 autocomplete="off"
 spellcheck="false"
 autofocus
 placeholder="Filter headings"
 aria-label="Filter headings">
</div>

```

```

<div class="SelectMenu-list SelectMenu-list--borderless p-2" style="overscroll-behavior:
contain;" data-filterable-for="toc-filter-field" data-filterable-type="substring">
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 text-emphasized" style="-webkit-box-orient: vertical; padding-left: 12px;"
data-action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating
_url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md', 'user_id': nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#ungraded-
lab-optional-build-train-and-deploy-an-xgboost-model-on-cloud-ai-platform">Ungraded Lab
(Optional): Build, train, and deploy an XGBoost model on Cloud AI Platform
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating
_url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md', 'user_id': nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#1-
overview">1: Overview
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating
_url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md', 'user_id': nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#2--setup-
your-environment">2 : Setup Your Environment
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating

```

```

_url";"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id";nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#3--
exploring-the-bigquery-dataset">3 : Exploring the BigQuery dataset
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type";"repository_toc_menu.click";"payload";
{"target";"entry";"repository_id";313476213,"originating
_url";"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id";nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#4--prepare-
data-for-training">4 : Prepare Data for Training
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type";"repository_toc_menu.click";"payload";
{"target";"entry";"repository_id";313476213,"originating
_url";"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id";nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#5--a-quick-
xgboost-primer">5 : A quick XGBoost primer
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type";"repository_toc_menu.click";"payload";
{"target";"entry";"repository_id";313476213,"originating
_url";"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id";nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#6--build-
train-and-evaluate-an-xgboost-model">6 : Build, train, and evaluate an XGBoost model
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type";"repository_toc_menu.click";"payload";
{"target";"entry";"repository_id";313476213,"originating
_url";"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md";"user_id";nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#7--deploy-
model-to-cloud-ai-platform">7 : Deploy model to Cloud AI Platform
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type";"repository_toc_menu.click";"payload";

```

```
{"target":"entry","repository_id":313476213,"originating_
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md","user_id":nu
ll}}" data-hydro-click-
hmac="a14c0aa3ad32d34b6c9db781b13136f74c526ea84c28e7a6683cc33604615c53" href="#8--
cleanup">8 : Cleanup
</div>
</div>
```

prevent recurring costs. Once you have it setup and you have your free credits, you can go to the [Cloud](#)

```
338 lines (224 sloc)

14.8 KB
```



► Click here to expand!

```
<div class="BtnGroup">
 <a data-permalink-href="/https-deeplearning-ai/machine-learning-engineering-for-
production-public/raw/34af5af3f5b09a52a1537292b0fe14a29ae16b8d/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md" href="/https-deeplearning-
ai/machine-learning-engineering-for-production-public/raw/main/course4/week1-ungraded-
labs/C4_W1_Optional_Lab_1_XGBoost_CAIP/C4_W1_Optional_Lab_1.md" id="raw-url"
group_item="true" data-view-component="true" class="js-permalink-replaceable-link Button--
secondary Button--small Button">
 Raw

```

## 4 : Prepare Data for Training

[Blame](#)

```
<div class="d-flex">
```

## 5 : A quick XGBoost primer

► Click here to expand!

```
<button class="btn-octicon btn-octicon-danger disabled tooltiped tooltiped-nw"
disabled
 aria-label="You must be signed in to make or propose changes" type="button">
 <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-trash">
<path d="M11 1.75V3h2.25a.75.75 0 0 1 0 1.5H2.75a.75.75 0 0 1 0-1.5H5V1.75C5 .784 5.784 0
6.75 0h2.5C10.216 0 11 .784 11 1.75ZM4.496 6.675l.66 6.6a.25.25 0 0 0 .249.225h5.19a.25.25
0 0 0 .249-.225l.66-6.6a.75.75 0 0 1 1.492.149l-.66 6.6A1.748 1.748 0 0 1 10.595 15h-
5.19a1.75 1.75 0 0 1-1.741-1.575l-.66-6.6a.75.75 0 1 1 1.492-.15ZM6.5 1.75V3h3V1.75a.25.25
0 0 0-.25-.25h-2.5a.25.25 0 0 0-.25.25Z"></path>
```

## 8 : Cleanup

► Click here to expand!

```
</div>
```



```
</main>
```

```
<footer class="footer width-full container-xl p-responsive" role="contentinfo">
```

## ■ Footer

---



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```

<nav aria-label='Footer' class="col-12 col-lg-8">
 <h3 class='sr-only' id='sr-footer-heading'>Footer navigation</h3>
 <ul class="list-style-none d-flex flex-wrap col-12 flex-justify-center flex-lg-justify-between mb-2 mb-lg-0" aria-labelledby='sr-footer-heading'>
 <li class="mr-3 mr-lg-0">Terms
 <li class="mr-3 mr-lg-0">Privacy
 <li class="mr-3 mr-lg-0"><a data-analytics-event="{"category":"Footer","action":"go to security","label":"text:security"}" href="https://github.com/security">Security
 <li class="mr-3 mr-lg-0">Status
 <li class="mr-3 mr-lg-0"><a data-ga-click="Footer, go to help, text:Docs" href="https://docs.github.com">Docs
 <li class="mr-3 mr-lg-0">Contact GitHub
 <li class="mr-3 mr-lg-0">Pricing
 <li class="mr-3 mr-lg-0">API
 <li class="mr-3 mr-lg-0">Training
 <li class="mr-3 mr-lg-0">Blog
 <a data-ga-click="Footer, go to about, text:about" href="https://github.com/about">About

</nav>

```

```
<template id="site-details-dialog">
```



```

<div class="Popover js-hovercard-content position-absolute" style="display: none; outline: none;" tabindex="0">

```

```
<template id="snippet-clipboard-copy-button">
```



```
</div>
```

```
<div id="js-global-screen-reader-notice" class="sr-only" aria-live="polite" ></div>
```